Every word, fully defined

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S1 · Satellite types & orbits

satellite
anything that orbits something bigger
anything that orbits something bigger. The Moon is Earth's natural satellite; everything we launch is an artificial one. *A ball on a string, swung so fast the string is gravity.*

orbit
falling and missing, forever
the closed path a satellite follows around a body. It's falling toward Earth and missing, forever. *Throw a ball sideways fast enough and the ground curves away as fast as it falls.*

altitude
height above the ground, not from center
height above the ground, not from Earth's center. GEO is 35,786 km *altitude*; its distance from the center is that plus Earth's 6,371 km radius.

LEO / MEO / GEO · also: low / medium / geosynchronous
200–2000 km · 2000–35,786 · 35,786
Low, Medium, Geosynchronous Earth Orbit: the three height bands. LEO 180–2,000 km, MEO 2,000–35,786 km, GEO 35,786 km. *Downstairs, middle floor, penthouse.*

orbital period (T)
time for one lap: 90 min / 12 h / 24 h
Time for one revolution: $T = 2\pi\sqrt{(a^3/\mu)}$. Depends only on a and the central body.

geosynchronous (GSO)
1-day lap, any tilt
an orbit whose period is exactly one day, at any tilt. The satellite returns to the same spot in the sky once a day but may drift north-south in between.

geostationary (GEO)
1-day lap + over equator, stays put
geosynchronous AND directly over the equator with no tilt, so it appears frozen in one spot. TV dishes point at these and never move. *Sync = same time every day. Stationary = never leaves.*

equator / equatorial plane
flat sheet through Earth's middle
the flat sheet through Earth's middle, perpendicular to its spin axis. The reference plane for Earth orbits.

sidereal day
true spin vs stars, 23 h 56 m
one true rotation of Earth relative to the stars: 23 h 56 min, not 24 h. That's the real period a GEO satellite matches. *The 4-minute difference is Earth also moving around the Sun.*

astronomical satellite · also: space telescope
looks out: Hubble, JWST
a telescope in space pointed away from Earth (Hubble, JWST). *Looks out.*

Earth observation (EO)
looks down: Landsat
cameras and radar pointed down at Earth for mapping, farming, disasters (Landsat). *Looks down.*

weather satellite
looks down from GEO: GOES
looks down at clouds and storms, usually from GEO so it watches the same region nonstop (GOES).

reconnaissance satellite · also: spy sat
looks down, military
a military spy satellite. Same idea as Earth observation, different customer and classified detail.

navigation satellite
broadcasts time, phone finds itself: GPS
broadcasts a precise time signal; your phone hears four of them and works out where it is (GPS, Galileo). *Tells you where you are.*

communication satellite · also: comms / relay
radio up, radio down: Starlink
a radio relay in the sky: receives a signal from one place on Earth and sends it back down somewhere else (Starlink, DirecTV). *A mirror for radio.*

biosatellite
carries living things: Laika
carries living things (plants, animals, cells) to study how they cope with space. Sputnik 2 carried Laika the dog in 1957.

space station
satellite people live in: ISS
a satellite people live in (ISS, China's Tiangong).

constellation
swarm so one is always overhead
A set of satellites in coordinated orbits that function as one system (GPS, Iridium, Starlink). Arranged in several orbital planes so coverage is continuous.

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S1 · Satellite types & orbits

footprint · also: coverage area
patch a sat sees; bigger when higher
The patch of Earth a satellite can see or communicate with at one moment. Grows with altitude. Also called: field of view (FOV) · swath

eclipse (orbital)
Earth blocks the Sun; batteries cover it
when Earth blocks the Sun from the satellite. LEO sats get ~15 a day; batteries have to cover them.

atomic oxygen / drag
LEO-only problems: eats plastic / slows you
a single O atom on its own, made when UV splits O_2. Craves a bond with anything, so it eats materials. Main gas from ~200–600 km. *A loose dog with no leash.*

S1 · Types of interplanetary spacecraft

interplanetary
between planets, orbiting the Sun
between planets. An interplanetary spacecraft leaves Earth's neighborhood entirely, as opposed to a satellite that stays in Earth orbit.

flyby
passes once, never captured
a spacecraft that passes a planet once at high speed and keeps going. It never slows down enough to be caught. *Driving past a house and photographing it out the window.*

orbiter
brakes into orbit, stays
a spacecraft that fires its engine to slow down at the planet so gravity captures it into orbit. *Pulling into the driveway and staying.*

atmospheric probe
dives into the air, short life
a small craft that dives into a planet's air on purpose, usually on a parachute or balloon, and reports until it's crushed or runs out of power (Galileo probe at Jupiter, Huygens at Titan).

lander
touches down, stays put
a craft that survives touchdown and sits in one place doing science (Viking, InSight, Philae).

rover
lander with wheels
a lander with wheels: it can drive to new spots (Sojourner, Spirit, Opportunity, Curiosity, Perseverance).

penetrator · also: impactor
crashes in on purpose
a craft designed to slam into the surface on purpose, either to bury sensors or to kick up material for another craft to analyze (Deep Impact hit a comet; LCROSS hit the Moon).

observatory spacecraft · also: space telescope
telescope platform, target = sky
a telescope platform whose target is the sky, not a planet (Hubble, JWST, Kepler).

communications spacecraft · also: relay
forwards messages for others
a relay whose job is passing messages between other spacecraft and Earth. Mars orbiters relay for the rovers; China's Queqiao relays for the far side of the Moon.

commitment ladder · also: mission types
flyby→orbiter→probe→lander→rover→penetrator
my phrase, not the instructor's: flyby → orbiter → atmospheric → lander → rover → penetrator, ordered by how much fuel, mass, and risk each needs, and how much science it returns.

delta-v (Δv) · also: velocity change · 'the fuel bill'
speed change = fuel bill
Total change in speed a maneuver or mission requires (km/s). Converts to propellant mass through the rocket equation. Also called: velocity change · 'the fuel bill'

braking burn · also: capture burn
orbit insertion · slow down so gravity grabs you
firing the engine against your direction of travel when you arrive, to slow down enough for the planet's gravity to hold you. A flyby skips this; an orbiter can't.

prograde / retrograde · also: posigrade / —
burn to speed up / slow down
prograde = firing so you speed up along your direction of motion; retrograde = firing to slow down. *Gas pedal vs brake.*

gravity assist · also: flyby
slingshot · steal speed from a moving planet
swinging close past a planet to steal a bit of its orbital speed, for free. New Horizons used Jupiter this way. *Grabbing a moving merry-go-round to fling yourself faster.*

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S1 · Types of interplanetary spacecraft

EDL · also: entry, descent, landing
7 minutes of terror
Entry, Descent, Landing: the sequence a lander or rover goes through from hitting the top of the atmosphere to touching down. On Mars it's ~7 minutes and fully automatic. *"Seven minutes of terror."*

C3 · also: characteristic / launch energy
how hard you throw it from Earth
a number for how hard you have to throw a spacecraft to leave Earth and reach a target, in km²/s². Bigger C3 = bigger rocket. Earth escape is C3 = 0; Mars needs ~10–15; Pluto direct needed far more.

Kuiper Belt
icy ring beyond Neptune, Pluto lives there
the ring of icy bodies beyond Neptune. Pluto is the biggest known member. New Horizons kept flying into it after Pluto.

comet vs asteroid
ice + dust with a tail / a rock
a chunk of ice and dust that grows a tail when it nears the Sun. Rosetta orbited one and dropped the Philae lander onto it.

S2 · Space vacuum conditions

vacuum
almost no gas: no pressure, no air
space with (almost) no air in it. Not "nothing"; just no gas pushing on things. *Like the inside of a sealed jar after you pump all the air out.*

ozone layer (O_3)
Earth's sunscreen, blocks UV < 0.3 µm
a thin band of a special oxygen (O_3) high in Earth's air that soaks up harsh sunlight before it reaches you. *Earth's sunscreen.*

ultraviolet (UV)
invisible high-energy light, sunburns
light that's a bit "bluer than blue," too high-energy for your eyes to see. It's what sunburns you. *The invisible part of sunlight that does damage.*

wavelength
size of one ripple; shorter = more energy
the size of one ripple of a light wave. Shorter wavelength = more energy = more damage. UV is shorter than visible light. *Small ripples hit harder.*

µm (micrometer) · also: micron
millionth of a meter; 0.3 µm = 300 nm
one millionth of a meter. 0.3 µm = 300 nm. A human hair is ~70 µm. *The caption said "millimeters": wrong by 1,000×.*

polymer · also: plastic
long molecule chains: Kapton, Teflon
any plastic, rubber, or similar material made of long repeating chains of molecules. Kapton, Teflon, epoxies, cable insulation. *Long molecular spaghetti.*

organic chemical bond · also: bond energy
carbon glue; UV photon beats it
the "glue" between carbon atoms that holds those chains together. Takes a certain amount of energy to break. *UV has more than that amount, so it snaps the glue.*

degradation / decomposition
material falling apart over time
the material slowly falling apart: going brittle, yellowing, cracking, flaking. *Like a garden hose left in the sun for 5 years.*

corrosion
chemical attack (rust); loose word for UV damage
chemical attack on a material, like rust on iron. The instructor used it loosely for UV damage; strictly, UV damage is bond-breaking, not corrosion. *Rust = corrosion. Sun-rotted plastic = degradation.*

UV photodegradation · also: UV degradation
photons snap polymer bonds
Breaking of chemical bonds in polymers by UV photons whose energy exceeds the bond energy (~4 eV at 300 nm vs ~3.6 eV for C–C). Also called: UV degradation

outgassing
gas leaking out of solids in vacuum
gas slowly leaking out of a solid material once it's in vacuum. *Like a new car's "new car smell": plastics releasing trapped stuff, but in space it lands on your camera lens.*

TML / CVCM
outgassing limits: < 1% / < 0.1%
Total Mass Loss and Collected Volatile Condensable Material: the two numbers a material gets from a standard outgassing test (heated in vacuum for 24 hours). TML is how much of its mass escapes as gas; CVCM is how much of that gas condenses on a cold plate nearby, which is what would fog a lens. Spacecraft materials must stay under 1% TML and 0.1% CVCM. *The 'how much will this thing sweat in space' report card.*

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S2 · Space vacuum conditions

bake-out
outgas it on the ground first
Heating components in a vacuum chamber before launch so they outgas on the ground instead of in orbit.

void
bubble inside a material
a tiny bubble or pocket inside a material left over from manufacturing, usually full of air. *A bubble in a cake.*

pressure difference
1 atm inside, 0 outside → cracks
one side pushing harder than the other. Air trapped in a void pushes out at 1 atmosphere; vacuum pushes back with nothing. *A balloon taken up a mountain swells for the same reason.*

clean room
filtered air, no specks on sensors
a sealed workspace with filtered air, bunny suits, and strict rules, so dust and skin flakes don't land on the spacecraft. *An operating theater for hardware.*

perturbation
small unwanted nudge
any small unwanted push, nudge, or wobble. A thruster puff, a solar panel unfolding, a wheel spinning up. *Someone bumping your desk.*

dissipation · also: damping
energy leaking away until motion stops
energy leaking away into the surroundings until the motion stops. *The reason a plucked guitar string goes quiet.*

Newton's 1st law · also: inertia
motion keeps going unless pushed
a thing keeps doing what it's doing (moving or sitting still) unless something pushes on it. *On Earth, air and friction are the "something." In space, there isn't one.*

vibration / damping
shaking / what makes it fade
a shaking back and forth / anything that makes it fade out. *Shock absorbers on a car are dampers.*

convection
heat carried by moving air; gone in vacuum
heat carried away by moving air or liquid: hot air rises, cool air replaces it. *Why a radiator heats a whole room.*

conduction
heat through touching solids
heat moving through a solid by touch, atom to atom. *A metal spoon getting hot in soup.*

radiation (heat)
heat glowing away; works in vacuum
heat glowing away as invisible light, needing no air at all. *Feeling a campfire on your face from a distance.* In vacuum, conduction + radiation are the only two left.

radiator
panel that glows heat into space
A panel that dumps the spacecraft's waste heat into space by radiation, the only heat exit in vacuum.

MLI · also: multilayer insulation
thermal blanket · crinkly gold wrap
Stacks of thin reflective film that slow radiative heat loss. The crinkly gold/silver wrapping on spacecraft. Also called: thermal blanket

equilibrium / hot spot
settled temperature / unspread heat
Thermal equilibrium is when heat has spread out so everything sits at the same temperature. A hot spot is a place where heat has piled up and not spread yet. On Earth convection evens hot spots out; in vacuum they stay hot until conduction or radiation slowly moves the heat. *A pan on the stove: the whole pan warms evenly in air; in vacuum only the spot over the flame would glow.*

electrical discharge · also: ESD
a spark
built-up charge suddenly jumping somewhere: a spark. *The zap from a doorknob after walking on carpet.*

spacecraft charging · also: surface charging · electrostatic discharge (ESD)
static builds, no air to bleed it, arcs
the spacecraft slowly collecting electric charge from space (sunlight knocks electrons off, charged particles stick). With no air to bleed it away, it builds until it sparks. *You on a carpet, forever, with no doorknob.*

passive vs active system
no power (coating) / powered (heater)
Passive: works with no power and no decisions, by its nature (a coating, a grounding strap, a radiator, a thermal blanket). Active: does something on command, using power (a heater, a reaction wheel, a thruster, a charge-bleeding device). Engineers prefer passive where possible because it can't fail by losing power. *A window shade vs. air conditioning.*

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S2 · Earth atmosphere

atmosphere
blanket of gas held by gravity
the blanket of gas held around a planet by gravity. Thick at the bottom, fading to nothing a few hundred km up. *Earth's air, all of it.*

troposphere
0–11 km, weather, colder up
bottom floor, 0–11 km. Weather, clouds, airliners. *"Tropo" = turning: the air churns here.*

stratosphere
11–50 km, ozone, warmer up
11–50 km. Calm, layered, holds the ozone layer, warms with height. *"Strato" = layered.*

mesosphere
50–90 km, coldest, meteors burn
50–90 km. The coldest layer; meteors burn up here. *"Meso" = middle.*

thermosphere
90–500 km, hot but empty, ISS here
90–500 km. Extremely hot by thermometer, almost no air. The ISS and most LEO satellites fly inside it. *"Thermo" = heat.*

exosphere
500 km+, fades to space
500 km and beyond. So thin that atoms can drift off into space. *"Exo" = outside.*

homosphere / heterosphere
stirred mix / sorted by weight
homo = the bottom ~100 km where gases are stirred into one uniform mix. Hetero = above that, where gases separate by weight. *Smoothie vs. a layered cocktail.*

turbopause · also: homopause
~100 km, mixing stops
The ~100 km boundary where turbulent mixing stops and diffusive separation begins. Also called: homopause

temperature vs heat
molecule speed / energy delivered
temperature = how fast molecules move. Heat = how much energy they can actually give you, which also depends on how many there are. *One sparkler is 1000 °C and won't warm a room.*

density / number density
how much stuff per volume (m^{-3})
how much stuff is packed into a space. Numerical density (m^{-3}) on the chart = how many molecules per cubic meter. *People per square meter in a crowd.*

log scale
each tick = 100,000×
the chart's vertical axis where each tick (10^{10}, 10^{15}, 10^{20}) is 100,000× the one below. Used when numbers span huge ranges. *Zooming out so a mountain and a pebble fit on one page.*

constituent · also: component
one gas in the mix
one of the gases the air is made of. *Ingredients.*

N_2 / O_2 / Ar / CO_2
78% / 21% / 0.9% / 0.04%
nitrogen gas (two N atoms), oxygen gas (two O atoms), argon. The three main ingredients of normal air: 78%, 21%, 1%.

O / N / H / He · also: atomic forms
single atoms, up high
single atoms of oxygen, nitrogen, hydrogen, and helium gas. The "atomic" forms. Up high, radiation breaks the paired molecules into these singles. *Loose Lego bricks instead of snapped-together pairs.*

atomic oxygen (AO)
lone O atom, 200–600 km, eats plastic
a single O atom on its own, made when UV splits O_2. Craves a bond with anything, so it eats materials. Main gas from ~200–600 km. *A loose dog with no leash.*

photodissociation
light splits $O_2 \rightarrow O + O$
Light breaking a molecule apart. $O_2 + UV \rightarrow O + O$.

anomalous O · also: hot oxygen
minor extra-hot O above 300 km
a tiny extra population of unusually energetic oxygen atoms that the chart's model adds above ~300 km. Minor; not something you need to remember beyond "it's still oxygen."

ionosphere
60–1000 km, charged, bends radio, delays GPS
~60–1,000 km, overlapping the upper layers: solar UV and X-rays ionize atoms, leaving free electrons that refract and reflect radio and delay GPS signals.

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S2 · Earth atmosphere

geocorona
hydrogen halo glowing in UV
The faint hydrogen halo of the exosphere, glowing in UV, extending tens of thousands of km.

VLEO / LEO
below ~250 km heavy drag / up to 2000
Very Low Earth Orbit (below ~250 km, heavy drag, short life) vs Low Earth Orbit (up to 2,000 km). The green line on the chart marks the split.

reactive
eager to bond, sticky
eager to bond chemically with other stuff. *Sticky.*

oxidation
oxygen bonding to a material; rust
oxygen bonding to a material and changing it. Rust is iron oxidizing. Atomic oxygen oxidizes spacecraft surfaces; here "corrosion" is the right word. *Rust, but in orbit, and fast.*

erosion
worn away bit by bit
material being worn away bit by bit. Atomic oxygen erodes polymers; the slide photo shows a surface roughened like sandblasted glass.

erosion yield / fluence
loss per atom / total atoms that hit
material being worn away bit by bit. Atomic oxygen erodes polymers; the slide photo shows a surface roughened like sandblasted glass.

kinetic energy / relative velocity
energy of motion / closing speed
kinetic energy = energy of motion. Relative velocity = how fast two things approach each other; it doesn't matter which one is "really" moving. The s/c at 7.5 km/s hitting a still atom is the same as the atom hitting a still s/c at 7.5 km/s.

eV (electron-volt)
tiny energy unit; bond ≈ few eV
a tiny unit of energy used for single atoms and bonds. A chemical bond is a few eV; an atomic-oxygen hit at 7.5 km/s carries ~5 eV. *Enough to snap the bond.*

ram / wake
front face / sheltered back
Ram: the side facing the direction of motion, which takes the atomic-oxygen and drag hits. Wake: the sheltered trailing side.

atmospheric drag · also: aerodynamic drag
thin air slows LEO sats; ISS reboosts
Resistance from residual air in LEO that slowly lowers the orbit. Grows with solar activity; the ISS reboosts to counter it. Also called: aerodynamic drag

Kapton · also: polyimide film
gold insulation, AO's victim
a gold-colored polymer film used everywhere on spacecraft for insulation. Famous for being eaten by atomic oxygen unless coated.

F10.7 / Ap · also: space weather indices
Sun activity; puffs up the air
two numbers on the chart title describing how active the Sun and Earth's magnetic field were for the model run. The atmosphere puffs up when the Sun is active. Ignore the values.

S3 · Orbit definition & parameters

orbital mechanics
math of where a s/c goes under gravity
the math of how things move under gravity: given where a spacecraft is and how fast it's going, where will it be later? *Ballistics, but the "ground" is a planet.*

celestial body
planet, moon, Sun, comet
any big natural object in space: planet, moon, Sun, asteroid, comet. The thing you orbit.

transfer orbit
the highway between two orbits
an orbit you use to get from one place to another, like Earth to Mars, rather than to stay. *The highway, not the parking lot.*

Classical Orbital Elements · also: Keplerian elements
the six · a e i Ω ω ν = the hoop's settings
The six numbers that completely describe an orbit: a, e, i, Ω, ω, ν. Five fix the shape and orientation; the sixth says where the body is. The hula hoop's six settings. Also called: Keplerian elements · orbital parameters · the six elements

semi-major axis (a) · also: half the major axis · 'size' of the orbit
size: (r_p + r_a)/2
half the longest width of the ellipse. Sets the orbit's size and its period. *"Semi" = half, "major axis" = the long way across.*

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S3 · Orbit definition & parameters

eccentricity (e) · also: 'shape' of the orbit
squash: 0 circle → 1 cigar → >1 open
a number from 0 up that says how squashed the ellipse is. 0 = circle, near 1 = very stretched, 1 or more = doesn't close. *"Eccentric" literally means off-center.*

perigee / apogee · also: periapsis / apoapsis
closest, fastest / farthest, slowest
closest and farthest points of an orbit around Earth, measured from Earth's center. *"Peri" = near, "apo" = far, "gee" = Earth.*

perihelion / aphelion
same, around the Sun
The closest and farthest points of an orbit around the Sun. Same idea as perigee and apogee, with the suffix changed for the body: -gee for Earth, -helion for the Sun, -lune for the Moon, -apsis for anything. Earth's own perihelion is in early January, about 147 million km from the Sun; aphelion in early July, about 152 million km. *Perigee and apogee, Sun edition.*

apsis (apsides)
either extreme point
either of those two extreme points. The line joining them is the long axis of the ellipse.

line of apsides · also: major axis
perigee↔apogee line
The straight line through periapsis and apoapsis, passing through the central body. Its half-length is a. Also called: major axis

inclination (i) · also: tilt
tilt vs equator; 0° flat, 90° polar
the tilt angle between the orbit's plane and the reference plane (the equator, for Earth). 0° flat, 90° over the poles.

ecliptic
plane of Earth's orbit; reference for Sun orbits
the flat sheet Earth's orbit around the Sun lies in. The reference plane for orbits around the Sun. *Not the same as the equator; they're tilted 23.4° apart.*

plane of reference
what you measure tilt from
the flat sheet you measure tilt from. Equator for Earth orbits, ecliptic for Sun orbits.

vernal equinox (♈) · also: First Point of Aries
reference direction · fixed zero direction in space
a fixed "12 o'clock" direction in space to measure angles from: the direction to the Sun on the first day of spring. The ♈ symbol (Aries) marks it on diagrams. *North on a compass, but for space.*

ascending node
hoop crosses equator going north
the two points where a tilted orbit pierces the reference plane. The ascending node is the one where the satellite is crossing from south to north. *Where the hoop pokes up through the table.*

line of nodes
where orbital plane cuts equator
The line where the orbital plane cuts the reference plane; it joins the ascending and descending nodes.

RAAN (Ω) · also: longitude of ascending node
swivel: ♈ → ascending node
Angle, measured in the reference plane from the vernal equinox (♈), to the point where the orbit crosses the plane heading north. Also called: longitude of the ascending node · 'swivel'

argument of periapsis (ω) · also: argument of perigee
spin in plane: node → perigee
the angle, measured in the orbit's own plane, from the ascending node to perigee. The orbit's spin-in-place. *"Argument" is old math-speak for "angle." Nobody is arguing.*

true anomaly (ν)
where the bead is now
the angle from perigee to where the spacecraft is right now. *"Anomaly" is old astronomy-speak for "angle along the orbit." Nothing is wrong.*

epoch
timestamp the elements are valid for
The timestamp the elements are valid for. Elements drift, so a set of COEs always comes with an epoch.

orbital plane
flat sheet the orbit lies in; passes through center
The flat sheet the orbit lies in. Always passes through the center of the body being orbited. Why GEO must be over the equator.

circular / elliptical / hyperbolic
e = 0 / 0–1 / > 1
orbit shapes by e: circle (e = 0), ellipse (0 1, an open swing-past like a flyby).

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S3 · Orbit definition & parameters

polar / equatorial orbit
i = 90° / i = 0°
the imaginary line around Earth's middle. A geostationary satellite has to sit above it; a satellite above Texas can't hover there because Earth's spin would carry Texas away underneath.

geometry (of an orbit)
size + shape + orientation
The orbit's size, shape and orientation taken together: a, e, i, Ω and ω. Everything about the orbit except where the spacecraft currently is on it (that's ν). Two spacecraft can share one geometry and be at different points. *The shape and tilt of the hoop, ignoring the bead.*

vis-viva · also: orbital energy equation
$v^2 = \mu(2/r - 1/a)$
$v^2 = \mu(2/r - 1/a)$. Speed anywhere on any orbit from distance and semi-major axis. Gives circular speed (a = r) and escape speed (a → ∞). Also called: orbital energy equation

μ (gravitational parameter) · also: GM
Earth 398,600 km^3/s^2
The body's mass times the gravitational constant, in one number: Earth μ = 398,600 km^3/s^2. Shows up in every orbit formula. Also called: GM · 'mu'

S3 · Earth orbits & constellations

Earth orbit · also: geocentric
goes around Earth, not the Sun
~200–2,000 km altitude; period 1.5–2 h. ISS, Hubble, Starlink, most Earth imaging.

rotation period
Earth spins once: 23 h 56 m
how long one full trip around takes. LEO ~90 min, GPS 12 h, GEO 24 h. Higher = longer. *Bigger circle, slower runner.*

altitude vs distance from center
GEO 35,786 alt = 42,164 from center
height above the ground, not from Earth's center. GEO is 35,786 km *altitude*; its distance from the center is that plus Earth's 6,371 km radius.

coverage / global coverage
what a sat sees / everything, always
Coverage is the patch of Earth a satellite can see or talk to at a given moment (its footprint). Global coverage means every point on Earth is covered all the time, which one satellite can never do and a constellation can. *A flashlight's circle of light vs. a stadium's floodlights.*

resolution
how much detail; closer = sharper
How much detail an instrument can make out: the smallest thing it can tell apart. Depends on the optics and on distance, so a LEO camera can see a car while a GEO weather camera sees whole storm systems. Higher orbit = bigger coverage but lower resolution. *Reading a street sign from 10 feet vs. from a mile.*

orbital plane (constellation) · also: Walker constellation
several rings so sats aren't bunched
The flat sheet the orbit lies in. Always passes through the center of the body being orbited. Why GEO must be over the equator.

revisit time
how long until same spot again
How long until a satellite (or constellation) sees the same spot again. Shorter with more satellites or higher altitude.

GPS / GLONASS / Galileo / BeiDou
US / Russia / EU / China nav constellations
Global Positioning System, the US navigation constellation at ~20,200 km in MEO.

Iridium
66 sats, 780 km, sat phones anywhere
A company and its satellite constellation: 66 active satellites at about 780 km in six near-polar orbital planes, giving satellite-phone and data coverage everywhere on Earth including the poles and oceans. Named for element 77 because the original plan had 77 satellites. *The swarm that makes a sat phone work in the middle of the ocean.*

data relay / transfer time
sat → ground; higher = more delay
Data relay is passing information from one spacecraft to another or down to Earth, often through a dedicated relay satellite. Transfer time is how long the signal takes; radio travels at light speed, so it grows with distance: ~0.12 s one way to GEO, 4–24 minutes to Mars. Higher orbits mean longer delay. *A game of telephone across the solar system, with real lag.*

trade-off · also: compromise
better coverage costs detail + delay
Giving up some of one good thing to get more of another, because you can't have all of both. Orbit choice is a trade-off: higher gets you coverage and costs you detail and delay; lower gets you detail and costs you coverage and lifetime. Engineering is mostly choosing trade-offs on purpose. *A big truck vs. a sports car: you don't get both.*

Every word, fully defined

Same order as the word bank. Bold = the term · grey = what else it's called · orange = the short line you write in the notebook · then the full meaning, with the picture in italics.

S3 · Earth orbits & constellations

sun-synchronous (SSO)
polar-ish LEO, same local sun time
A near-polar LEO (~98° inclination) whose plane precesses once a year (via J_2) so it passes over any spot at the same local solar time. Consistent lighting for cameras.

J_2 / nodal precession
Earth's bulge swivels Ω; makes SSO work
Earth's equatorial bulge slowly swivels an inclined orbit's Ω. Engineers exploit it to make sun-synchronous orbits.

S3 · Transfers & flybys

Lambert's problem
2 points + a time → the orbit + Δv
the classic "find the orbit" puzzle: given a start point, an end point, and a travel time, what orbit connects them? Named after Johann Lambert, an 18th-century mathematician. *The "can I get there by Tuesday?" calculator.*

input / constraint
what you feed in / limit you can't break
what you feed into a calculation / a limit you can't break (rocket size, fuel mass, time). *Repeat.*

conic / patched conics · also: conic approximation
stitched 2-body pieces of a trajectory
Approximating an interplanetary trajectory as a chain of two-body conics: Earth-centered hyperbola → Sun-centered ellipse → target-centered hyperbola, 'patched' at the sphere-of-influence boundaries. Also called: conic approximation

sphere of influence (SOI)
where a planet's gravity wins; Earth ≈ 925,000 km
The region around a planet where its gravity dominates the Sun's for trajectory purposes. Earth's ≈ 925,000 km.

Hohmann transfer · also: minimum-energy two-impulse transfer
lazy half-ellipse, 180°, 2 burns, cheapest
the half-ellipse path that touches both the starting and ending orbit, with departure and arrival 180° apart and two burns. Cheapest in fuel. Worked out by Walter Hohmann, a German engineer, in 1925. *The lazy throw.*

tangent
just touching; why burns are small
just touching, not crossing. The Hohmann ellipse is tangent to both circles, which is why the burns are small: you're already moving along the right line.

optimal / efficient
least Δv, not fastest
in this lecture, always means "least Δv," i.e. least fuel. Not fastest.

time of flight (TOF) · also: transfer time
Hohmann = half the ellipse's period
how long the trip takes. For a Hohmann it's half the ellipse's period. *Your Section 1 quiz question.*

launch window / synodic period
lineup repeats; Mars every 26 months
the stretch of days when the planets are lined up right for a Hohmann. Earth–Mars: about every 26 months. *Repeat from S1 primer.*

phase angle · also: lead angle
how far ahead target must be: Mars ~44°
how far ahead the target planet must be when you launch so it's there when you arrive. Mars: ~44° ahead. Not on the slide; it's why the window exists.

impulsive burn · also: impulsive maneuver
engine firing treated as instant
An engine firing treated as instantaneous. Hohmann assumes two of them. Also called: impulsive maneuver

circularization burn · also: apogee kick
2nd burn rounds the ellipse
The second burn of a Hohmann: at apoapsis, speed up to round the ellipse into the target circle. Also called: apogee kick

plane change · also: inclination change
tilting the orbit; very expensive
A burn to tilt the orbital plane. Very expensive: $\Delta v = 2v \cdot \sin(\Delta i/2)$. Done at apogee where v is small. Also called: inclination change

bi-elliptic transfer
3-burn alternative, only for huge jumps
A three-burn alternative that goes far out and comes back. Beats Hohmann only for very large radius ratios. Named so you recognize it.

gravity field
region where the planet's pull matters
the region around a planet where its pull matters. It's what bends the flyby path.

Every word, fully defined

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S3 · Transfers & flybys

relative velocity
speed as seen from Mars vs from Sun
your speed as seen from something else (Mars, say) rather than from the Sun. The flyby doesn't change your speed relative to Mars, only relative to the Sun. *Repeat.*

vector / magnitude / direction
arrow / its length / which way
an arrow: a quantity with both a size and a direction. Velocity is a vector. The slide's arrow pictures are vector addition: spacecraft-relative-to-Mars plus Mars-relative-to-Sun equals spacecraft-relative-to-Sun. *Two pushes added tip to tail.*

v∞ (hyperbolic excess velocity)
speed relative to planet, far away
Your speed relative to a planet when far from it, before capture or after escape. The 'in' and 'out' speed of a flyby, equal in magnitude in the planet's frame.

turning angle (δ) · also: deflection angle
how much the flyby bends you
How much a flyby bends your direction. Larger for slower approaches and closer passes. Also called: deflection angle

orbital energy · also: specific orbital energy
speed + height budget; bigger = bigger a
the combined speed-plus-height "budget" of an orbit. Fixed unless you burn or do a flyby. Bigger budget = bigger orbit.

aerobraking
use atmosphere to shed speed instead of fuel
Using a planet's upper atmosphere to shed speed instead of fuel (gradually, over many passes = aerobraking; in one pass = aerocapture, not yet flown).

Tsiolkovsky rocket equation · also: ideal rocket equation
$\Delta v = Isp \cdot g_0 \cdot \ln(m_0/m_f)$
$\Delta v = I_{sp} \cdot g_0 \cdot \ln(m_0/m_f)$. Links fuel fraction to Δv; why fast transfers are impossible. Coming in the propulsion section. Also called: ideal rocket equation

Rosetta / 67P / ESA
comet mission 2004→2014, 4 flybys
the European Space Agency's comet mission: launched 2004, four flybys (Earth, Mars, Earth, Earth), arrived at comet 67P/Churyumov–Gerasimenko in 2014, dropped the Philae lander. *Rosetta and ESA are repeats; 67P is new.*

deep space
beyond the Moon; outer planets
far beyond Earth and the Moon; the outer planets and beyond.